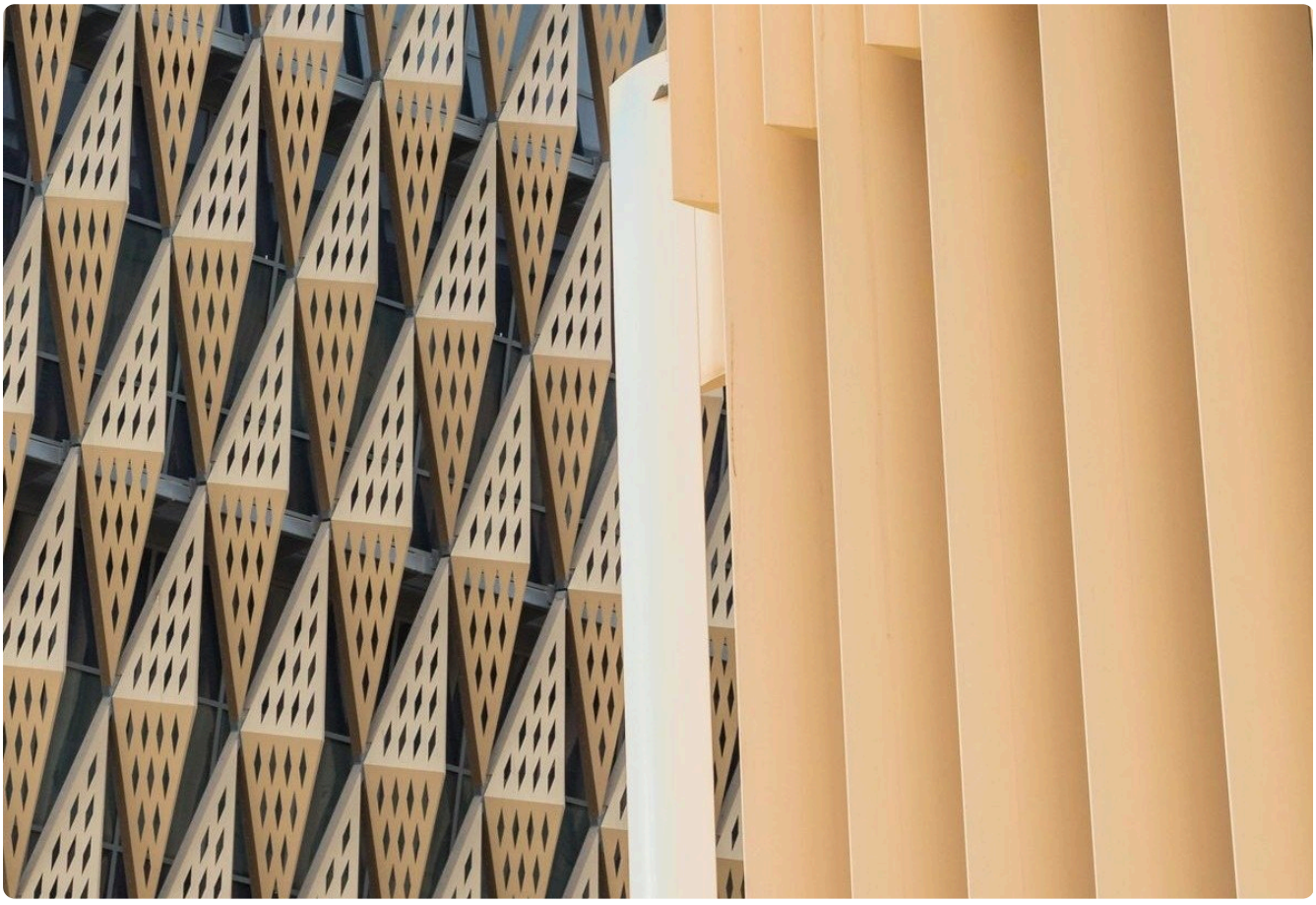
DESIGN COMPUTED · FREE GUIDE

Can AI design a building? The four checks every design must pass

The free guide to the reel on AI and architecture: what AI does well today, what it cannot check on its own, and a simple four-check routine to test any AI-generated design.



How it works

Generative AI tools can produce thousands of building images or layout options in minutes. That is a real change for the early, creative part of design: more options, faster.

But a picture of a building is not a building. A real design has to stand up, keep the rain out, follow the building code and fit the budget. A generated image does not know any of that unless a person or a specialised tool checks it.

Research reviews of generative AI in building design agree on the split: AI is useful for generating and screening options, and early work is showing progress on energy-code checks. Structural code compliance is still described as a goal for future tools, so it stays with licensed engineers and official review.

The four checks

Stand up	Can the structure carry its own weight, people, wind and, where relevant, earthquakes? Needs an engineer and calculations.
Keep the rain out	Does the envelope have a real roof slope, drainage, joints and layers that can be built?
Meet the code	Fire exits, stairs, accessibility, energy: each country and city has its own rules.
Fit the budget	Can it be built for the money, with materials and labour available?

Built examples of computation helping design

Guggenheim Bilbao (1997)	Digital models turned a sculptural shape into panels a factory could make.
Sagrada Família, Barcelona	Digital tools help make Gaudí's complex geometry buildable.
Lotus Temple, Delhi (1986)	Engineers resolved the curves into geometry that could be calculated.
BIQ house, Hamburg (2013)	A pilot facade tested with monitoring, not only with renders.

Try it yourself

1. Generate five building images with any free AI image tool and pick your favourite.
2. Draw its plan on paper: where are the walls, the stairs, the doors?
3. Check 1: draw the path of the loads from the roof to the ground. Where does a column or wall have to be?
4. Check 2: mark where rain falls and where it runs off. Where would water enter?
5. Check 3: look up the basic rules for stairs and exits in your country. Does your plan follow them?
6. Check 4: estimate the floor area and multiply by a typical local cost per square metre.
7. Write one line: which checks did the image pass, which did it fail, and what would you change?

The rule to remember

Next time AI gives you a pretty picture, ask which check it passed.

Trusting the picture	A render shows light and style, not structure or cost.
Skipping the engineer	Structural safety is a professional responsibility.
Using AI only at the end	Use it for options early, and keep the checks throughout.

Go further

Exploring Gen-AI applications in building research and industry: a review, arXiv
https://arxiv.org/pdf/2410.01098

Copilot for constraint-driven generation of architectural design evaluations, University of Stuttgart
https://www.itech.uni-stuttgart.de/itech-thesis-projects/2025-copilot-for-constraint-driven-generation-of-architectural-design-evaluations-and-suggestions/

Credits
Photos: Pexels, free licence (no credit required).
Facts: arXiv review, University of Stuttgart (iTECH).

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